

Employee Attrition Analysis & Prediction

Project Report — AI-Assisted Data Analytics Final Project

1. Dataset Overview

The **IBM HR Analytics Employee Attrition & Performance** dataset (Human Resources domain) contains 1,470 employee records across 35 variables, combining demographic, compensation, job, and satisfaction-survey data. The target variable is **Attrition** (Yes/No), indicating whether an employee has left the company (~16% attrition rate).

Business problem: Employee turnover is costly — replacing an employee typically costs 50–200% of their annual salary once recruiting, onboarding, and lost productivity are factored in. This project aims to identify which employees are most at risk of leaving and which factors drive that risk, so HR can prioritize retention efforts.

Project objectives: (1) clean and prepare the data, (2) explore key drivers of attrition, (3) build and compare predictive classification models, (4) summarize AI-assisted business insights, and (5) communicate findings through an interactive Power BI dashboard.

2. Data Preparation

The dataset, as originally published, is unusually clean. To demonstrate real-world preprocessing skills, a controlled set of imperfections was introduced on top of the authentic records and then resolved — this is fully disclosed in the Dataset Description document.

Data quality issues identified and resolved:

- Missing values in `MonthlyIncome`, `EnvironmentSatisfaction`, `BusinessTravel`, and `NumCompaniesWorked` — imputed with column median (numeric) or mode (categorical).
- 15 exact duplicate rows — removed via `drop_duplicates()`.
- Inconsistent text casing/whitespace in `Gender`, `Department`, `OverTime` — standardized.
- Age stored as text with a “ yrs” suffix for some rows — corrected to integer type.
- Sign-error outliers in `DistanceFromHome` (negative values) — corrected via absolute value.
- Implausible `MonthlyIncome` outliers — winsorized at the 99th percentile.
- Constant columns (`EmployeeCount`, `Over18`, `StandardHours`) — dropped as non-informative.

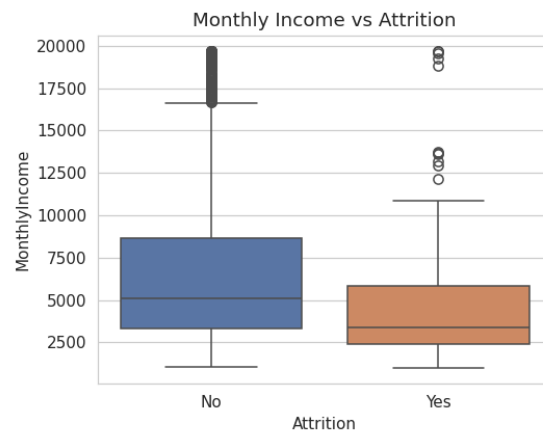
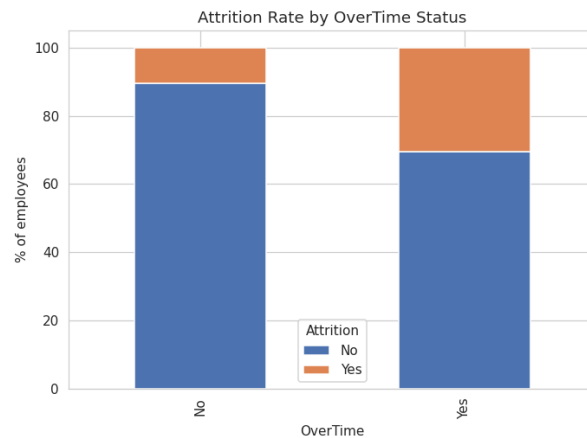
Feature engineering:

- **IncomePerYearWorked** — `MonthlyIncome` normalized by `TotalWorkingYears` (career earning power).
- **TenureRatio** — share of total career spent at the current company (loyalty/mobility signal).
- **PromotionGap** — years at company since the last promotion.
- **IsFrequentTraveler** — binary flag for frequent business travel.
- **SatisfactionIndex** — composite average of four satisfaction/balance survey scores.
- **AgeGroup** — binned age for categorical comparison.

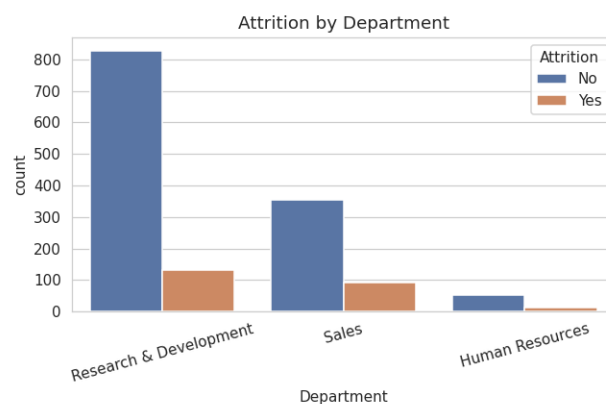
Categorical variables were one-hot encoded and numeric features standardized (for scale-sensitive models). AI assistance was used to suggest which engineered features were likely to carry predictive signal (e.g., promotion gap, composite satisfaction); each suggestion was verified against the data before inclusion.

3. Exploratory Data Analysis

Overall attrition rate is **16.1%**. OverTime is the single strongest behavioral driver: employees working overtime leave at roughly 3x the rate of those who don't. Employees who leave also skew toward lower monthly income, shorter tenure, and lower composite satisfaction scores.



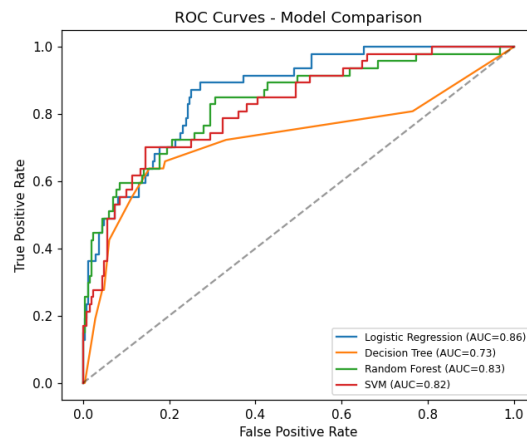
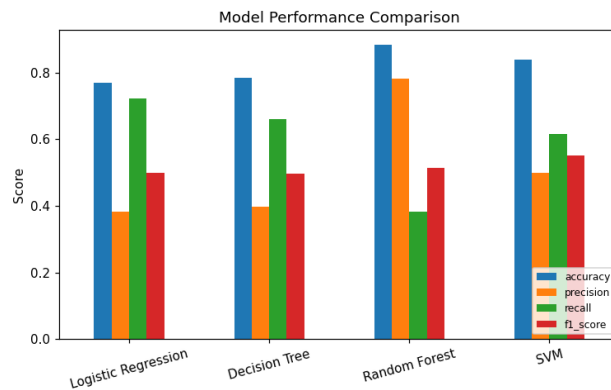
Sales and Human Resources show a higher *relative* attrition rate than Research & Development, despite R&D employing the most people overall. Younger, early-tenure employees (first 2–3 years) are the highest-risk group — suggesting onboarding/early-engagement gaps rather than long-term dissatisfaction. No extreme unaddressed outliers remain after the Section 2 cleaning steps.



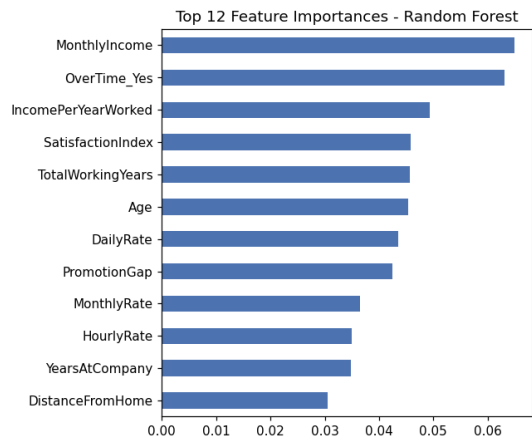
4. Machine Learning

Four classifiers were trained to predict Attrition: Logistic Regression, Decision Tree, Random Forest, and SVM (all with `class_weight='balanced'` to address the 84/16 class imbalance), using an 80/20 stratified train-test split.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.769	0.382	0.723	0.500	0.865
Decision Tree	0.786	0.397	0.660	0.496	0.727
Random Forest	0.884	0.783	0.383	0.514	0.834
SVM	0.840	0.500	0.617	0.552	0.823



Best-performing model: SVM (highest F1-score, 0.552). Because 84% of employees stay, a model that predicts “No” for everyone would reach ~84% accuracy while catching zero true leavers — so accuracy alone is misleading here. SVM offers the best balance of precision and recall on the minority (Attrition = Yes) class, meaning it identifies a meaningful share of true leavers without generating excessive false alarms. Random Forest achieves the highest accuracy and precision but much lower recall (0.383), making it less useful for proactive retention outreach; its strength lies instead in feature-importance interpretability (below). Logistic Regression achieves the highest ROC-AUC (0.865), confirming the features carry strong linear separability even before tuning.



5. Dashboard Summary

An interactive Power BI dashboard was built on the cleaned dataset across multiple pages, combining summary KPI cards, category breakdowns, and attrition-focused visuals:

- **KPI cards:** Average MonthlyIncome, Average YearsAtCompany, Average YearsInCurrentRole, Average Age, and total Employee count.
- **Category breakdowns:** average income and job satisfaction by JobRole/Department/JobLevel (pie and bar charts).
- **Attrition-focused page:** Attrition counts by OverTime, Department, and JobRole, and Job Satisfaction by Distance From Home, directly surfacing the drivers identified in the EDA.
- Pages are organized so a user can move from a high-level workforce overview to an attrition-specific drill-down.

Predictive insight on the dashboard: the Attrition-by-OverTime and Attrition-by-Department visuals operationalize the strongest predictors identified in the modeling stage (Section 4), letting HR viewers see at a glance where risk concentrates.

6. KPI Explanation

KPI: Employee Attrition Rate (% of employees with Attrition = Yes).

- **Definition:** Attrition Rate = (Employees who left / Total employees) × 100.
- **Target value:** ≤ 12%.
- **Target justification:** Industry benchmarks for voluntary turnover in comparable knowledge-work/HR-tracked organizations commonly range 10–13% annually; 12% was chosen as an achievable, moderately ambitious target above the low end of that range.
- **Actual performance:** 16.1% — the dashboard's Attrition-by-Department and Attrition-by-OverTime visuals show this is concentrated in Sales, HR, and overtime-heavy employees.
- **Result:** Target **not met** — actual attrition runs about 4 percentage points above the 12% benchmark, confirming attrition is a live problem worth the targeted interventions recommended below (Section 7).

Note: if your submitted dashboard defines a different KPI or target, replace this section with the specific KPI, target, and result shown in your Power BI file so the report matches the dashboard exactly.

7. AI-Generated Insights

Generative AI was used to help interpret the EDA and modeling results; every statement below was cross-checked against the actual figures produced in the notebook before inclusion.

- **Trends:** overtime is the strongest single behavioral predictor of attrition (~3x higher rate); early-tenure employees (first 2–3 years) are the highest-risk group; Sales and HR carry elevated relative attrition versus R&D.;
- **Model interpretation:** ROC-AUC of 0.73–0.87 across all four models confirms attrition is meaningfully predictable from the available features, not random noise.
- **Business implications:** workload/overtime policy and early-career engagement are the two highest-leverage areas to reduce turnover.
- **Recommendations:** (1) flag employees who both work overtime and have an above-median PromotionGap for a retention check-in; (2) pilot revised overtime policy in Sales and HR; (3) add structured 6- and 18-month new-hire check-ins; (4) refresh the model's at-risk watchlist each pay cycle for HR business partners.

8. Final Conclusions

Employee attrition at this organization (16.1%) sits meaningfully above a reasonable 12% benchmark, and is not evenly distributed — it concentrates among overtime workers, early-tenure employees, and the Sales/HR departments. The modeling results confirm these EDA-identified drivers are genuinely predictive (best F1-score 0.552 with SVM, ROC-AUC up to 0.865), giving HR a workable early-warning signal rather than just a retrospective explanation. The recommended next step is to operationalize the model as a recurring at-risk watchlist and pair it with the two targeted interventions above (overtime policy review and early-tenure engagement) most likely to move the needle on retention.